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Buckle device

Abstract

In a buckle device, a buckle body is held in an upper part of a boot, and a lower part of the boot is supported from the lower side by a frame and a protector. Here, an upper left inclination surface of the boot and the lower left inclination surface of the protector are inclined in a downward direction as being directed inward in the right-left direction of the boot. Therefore, in a case in which a load is input to the boot from the upper side, it is possible to apply a displacement force to the boot inward in the right-left direction of the protector, and to suppress the downward displacement of the boot with respect to the frame and the protector.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2023-022015 filed on Feb. 15, 2023, the disclosure of which is incorporated by reference herein.

BACKGROUND

Technical Field

(2) The present invention relates to a buckle device in which a holding member holds a buckle body and a support supports the holding member.

Related Art

(3) In the buckle device disclosed in Japanese Patent Application Laid-Open (JP-A) No. 2020-66332, a boot holds a buckle body and an anchor supports the boot.

(4) Here, in the buckle device, it is preferable that the displacement of the boot to the anchor side can be suppressed.

SUMMARY

(5) In view of such circumstances, an object of the invention is to obtain a buckle device capable of suppressing displacement of a holding member to a support side.

(6) According to a first aspect of the invention, a buckle body configured to engage with a tongue provided at a webbing attached to an occupant; a holding member that holds the buckle body and is provided with a support target surface; a support that is provided with a support surface and supports the holding member in a manner such that the support surface supports the support target surface; and an inclination part that is provided at at least one of the support target surface or the support surface and is inclined in a direction from a buckle body side toward a support side as being directed inward in a holding-member thickness direction.

(7) In the buckle device in the first aspect of the invention, the buckle body is configured to engage with the tongue provided at the webbing attached to the occupant. The holding member holds the buckle body, the support surface of the support supports the support target surface of the holding member, and the support supports the holding member.

(8) Here, the inclination part of at least one of the support target surface or the support surface is inclined in a direction from the buckle body side toward the support side as being directed inward in the holding-member thickness direction. Therefore, it is possible to suppress displacement of the holding member to the support side.

(9) In a buckle device according to a second aspect of the invention, in the buckle device according to the first aspect of the invention, the inclination part is provided at the support target surface and the support surface.

(10) In the buckle device according to the second aspect of the invention, the inclination part is provided at the support target surface and the support surface. Therefore, it is possible to effectively suppress the displacement of the holding member to the support side.

(11) In a buckle device according to a third aspect of the invention, in the buckle device according to the second aspect of the invention, the inclination part at the support surface supports the inclination part at the support target surface.

(12) In the buckle device according to the third aspect of the invention, the inclination part at the support surface supports the inclination part at the support target surface. Therefore, it is possible to effectively suppress the displacement of the holding member to the support side.

(13) In a buckle device according to a fourth aspect of the invention, in the buckle device according to any one of the first to third aspects of the invention, the inclination part is provided on both sides in the holding-member thickness direction.

(14) In the buckle device according to the fourth aspect of the invention, the inclination part is provided on both sides in the holding-member thickness direction. Therefore, it is possible to effectively suppress the displacement of the holding member to the support side.

(15) In a buckle device according to a fifth aspect of the invention, in the buckle device according to any one of the first to fourth aspects of the invention, a plurality of inclination parts are provided on one side in the holding-member thickness direction.

(16) In the buckle device according to the fifth aspect of the invention, the plurality of the inclination parts are provided on one side in the holding-member thickness direction. Therefore, it is possible to effectively suppress the displacement of the holding member to the support side.

(17) In a buckle device according to a sixth aspect of the invention, in the buckle device according to any one of the first to fifth aspects of the invention, the support target surface and the support surface are not provided on both sides of the inclination part in a holding-member circumferential direction.

(18) In the buckle device according to the sixth aspect of the invention, the support target surface and the support surface are not provided on both sides of the inclination part in the holding-member circumferential direction. Therefore, even in a case in which the support target surface and the support surface are not provided on both sides of the inclination part in the holding-member circumferential direction, it is possible to suppress the displacement of the holding member to the support side by the inclination part.

(19) A buckle device according to a seventh aspect of the invention in the buckle device according to any one of the first to sixth aspects of the invention further includes an accommodation part that is provided at the holding member and accommodates the support, an exposure hole that is provided at a side wall of the accommodation part and exposes the support, and a covering part that is provided at the side wall and covers the support.

(20) In the buckle device according to the seventh aspect of the invention, the support is accommodated in the accommodation part of the holding member, and the exposure hole of the side wall of the accommodation part exposes the support.

(21) Here, the covering part of the side wall of the accommodation part covers the support. Therefore, in a case in which the holding member is displaced to the support side, it is possible to suppress an occurrence of a situation in which the support comes out from the exposure hole of the side wall of the accommodation part, by the covering part, and to suppress the displacement of the holding member to the support side.

(22) A buckle device according to an eighth aspect of the invention includes a buckle body configured to engage with a tongue provided at a webbing attached to an occupant; a holding member that holds the buckle body and is provided with an accommodation part; a support that is accommodated in the accommodation part and supports the holding member; an exposure hole that is provided at a side wall of the accommodation part and exposes the support; and a covering part that is provided at the side wall and covers the support.

(23) In the buckle device in the eighth aspect of the invention, the buckle body is configured to engage with the tongue provided at the webbing attached to the occupant. The holding member holds the buckle body, the accommodation part of the holding member accommodates the holding member, and the support supports the holding member. The exposure hole of the side wall of the accommodation part exposes the support.

(24) Here, the covering part of the side wall of the accommodation part covers the support. Therefore, in a case in which the holding member is displaced to the support side, it is possible to suppress an occurrence of a situation in which the support comes out from the exposure hole of the side wall of the accommodation part, by the covering part, and to suppress the displacement of the holding member to the support side.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiments of the present invention will be described in detail based on the following figures, wherein:

(2) FIG. 1 is a cross-sectional view illustrating a buckle device according to an embodiment of the invention as viewed from a rear;

(3) FIG. 2 is a rear view illustrating an inside of the buckle device according to the embodiment of the invention as viewed from the rear;

(4) FIG. 3 is a front view illustrating the buckle device according to the embodiment of the invention as viewed from a front;

(5) FIG. 4 is a front view illustrating a frame, a boot, and the like of the buckle device according to the embodiment of the invention as viewed from the front;

- (6) FIG. 5A is a perspective view illustrating the buckle device according to the embodiment of the invention as viewed from a diagonally front right;
- (7) FIG. 5B is a perspective view illustrating the buckle device according to the embodiment of the invention as viewed from a diagonally bottom right;
- (8) FIG. 5C is a perspective view illustrating the buckle device according to the embodiment of the invention as viewed from a diagonally rear bottom;
- (9) FIG. 6A is a perspective view illustrating a protector and the like of the buckle device according to the embodiment of the invention as viewed from a diagonally front right;
- (10) FIG. 6B is a perspective view illustrating the protector and the like of the buckle device according to the embodiment of the invention as viewed from a diagonally front left;
- (11) FIG. 7A is a perspective view illustrating the protector and the like of the buckle device according to the embodiment of the invention as viewed from a diagonally rear right;
- (12) FIG. 7B is a perspective view illustrating the protector and the like of the buckle device according to the embodiment of the invention as viewed from a rear side;
- (13) FIG. 8A is a view illustrating the boot and the protector of the buckle device according to the embodiment of the invention as viewed from a bottom;
- (14) FIG. 8B is a view illustrating the boot of the buckle device according to the embodiment of the invention as viewed from the bottom;
- (15) FIG. 9A is a view illustrating the boot and the protector of the buckle device according to the embodiment of the invention as viewed from a diagonally front bottom; and
- (16) FIG. 9B is a view illustrating the boot of the buckle device according to the embodiment of the invention as viewed from the diagonally front bottom.

DETAILED DESCRIPTION

- (17) FIG. 1 is a cross-sectional view illustrating a buckle device **10** according to an embodiment of the invention as viewed from the rear. In the drawings, the front of the buckle device **10** is indicated by an arrow FR, the right (obverse side) of the buckle device **10** is indicated by an arrow RH, and the upper side of the buckle device **10** is indicated by an arrow UP.
- (18) The buckle device **10** according to the present embodiment constitutes a seat belt device **12** of a vehicle (automobile). The seat belt device **12** is applied to a seat (not illustrated) in a vehicle interior. The seat belt device **12** is provided with a winding device (not illustrated). The winding device is installed on the vehicle width direction outer side and the lower side of a seat rear part. In the winding device, a webbing **14** (see FIG. 1) having a long band shape is wound from the proximal end side. A webbing **14** is biased to the winding side to the winding device and pulled out upward from the winding device. The winding device is provided with a lock mechanism, and the lock mechanism locks pulling-out of the webbing **14** from the winding device in an emergency of the vehicle (in collision or the like).
- (19) The webbing **14** passes through a through anchor (not illustrated) movably on the distal end side with respect to the winding device. The through anchor is installed on the vehicle width direction outer side and the upper side of the seat rear part. An anchor (not illustrated) is fixed to the distal end part of the webbing **14**, and the anchor is installed on the vehicle width direction outer side and the lower side of the seat rear part. The webbing **14** movably passes through a tongue **16** (see FIG. 1) between the through anchor and the anchor.
- (20) The buckle device **10** is installed on the inner side and the vehicle width direction lower side of the seat rear part. The front, the right, and the upper side of the buckle device **10** are directed toward the front side or the rear side, and directed inward in the vehicle width direction, and directed upward, respectively.
- (21) As illustrated in FIG. 1, a buckle body **18** having a substantially rectangular parallelepiped shape is provided at an upper part of the buckle device **10**. The tongue **16** is engageable with the buckle body **18** from the upper side. The tongue **16** is engaged with the buckle body **18**, and the webbing **14** is attached to an occupant seated on the seat. As a result, a part (shoulder webbing) of

the webbing **14** between the through anchor and the tongue **16** is stretched obliquely from the shoulder part to the waist part (including the chest part) of the occupant, and a part (wrap webbing) of the webbing **14** between the tongue **16** and the anchor is stretched laterally to the waist part of the occupant. The engagement of the tongue **16** with the buckle body **18** is releasable. The engagement of the tongue **16** with the buckle body **18** is released to release the attachment of the webbing **14** to the occupant. The lower side part of the buckle body **18** gradually becomes smaller downward in the right-left direction.

(22) A buckle switch (not illustrated) as a detection unit is provided in the buckle body **18**. The buckle switch detects the engagement of the tongue **16** with the buckle body **18** and detects the attachment of the webbing **14** to the occupant. A plurality of buckle switches may be provided.

(23) A distal-end side part (upper side part) of a band-shaped belt **20** (webbing) as a coupling member is coupled to the lower part of the buckle body **18**. The belt **20** is made of, for example, the same material as the material of the webbing **14** and made into a woven fabric.

(24) A frame **22** that is made of metal as a support member constituting a support is provided in a lower part of the buckle device **10**, and the frame **22** is made to have a U-shaped plate shape in a cross section. A back plate **22A** is provided at a left part of the frame **22**. The frame **22** is fixed to a vehicle body side (for example, a rear part of a seat lower part) at a lower end part of the back plate **22A**. Leg plates **22B** and **22C** are provided at the front part and the rear part of the frame **22**, respectively. The leg plates **22B** and **22C** protrude rightward from the back plate **22A**. A hook protrusion **22D** having an L-shaped plate shape as a hook part is integrally formed at the right part of the lower surface of the leg plate **22B**. The hook protrusion **22D** protrudes downward and has a distal end part (lower end part) that protrudes leftward.

(25) A spool **24** that is made of metal and has a substantially cylindrical shape as a winding shaft passes through the leg plates **22B** and **22C** of the frame **22**. The spool **24** is supported by the frame **22**. The axial direction of the spool **24** is parallel to the front-rear direction, and the spool **24** is made rotatable about the central axis. A proximal-end side part (lower side part) of the belt **20** is coupled to the spool **24**. The belt **20** is wound around the spool **24** and pulled out upward from the left side of the spool **24**.

(26) A protector **26** (see FIGS. **6A**, **6B**, **7A**, and **7B**) that is made of resin as a covering member constituting the support is fitted into an upper part of the frame **22**. The protector **26** is made to have a U-shape in top view. The inside of the protector is opened to the right side, and the left wall, the front wall, and the rear wall of the protector **26** cover the back plate **22A**, the leg plate **22B**, and the leg plate **22C** of the frame **22**, respectively. The upper end part of the protector **26** is made to serve as an upper wall. The upper wall of the protector **26** protrudes to the outside of the protector **26** and is placed on the upper side of the frame **22**. A part of the front part and a part of the rear part of the protector **26** are made to have an inverted J-shape in cross section. The leg plate **22B** and the leg plate **22C** are fitted into these parts, respectively. The belt **20** is inserted into the protector **26**, and the protector **26** restricts the contact of the belt **20** with the frame **22**.

(27) A lower support surface **26B** that constitutes a support surface **26A** and has a substantially L-shape in top view is formed on the upper surfaces of the front part and the left part of the upper wall of the protector **26**. The lower support surface **26B** is disposed vertically in the up-down direction. A predetermined number (two in the embodiment) of support bases **26C** having a substantially trapezoidal columnar shape are formed at the left side part of the upper wall left part of the protector **26**. The predetermined number of support bases **26C** extend in the front-rear direction and are separated from each other. A lower-left inclined surface **26D** as an inclination part constituting the support surface **26A** is formed on the upper surface of the support base **26C**. The lower-left inclined surface **26D** is inclined in a downward direction as being directed to the right (the inner side of the protector **26** in the right-left direction). A support column **26E** having a substantially rectangular columnar shape is integrally formed at the right end part of the upper wall rear part of the protector **26**. The support column **26E** protrudes upward. A lower-right inclined

surface **26F** as the inclination part constituting the support surface **26A** is formed on the upper surface of the support column **26E**. The lower-right inclined surface **26F** is inclined in a downward direction as being directed to the left (the inner side of the protector **26** in the right-left direction).

(28) A case **28** (see FIGS. **3**, **5A**, **5B**, and **5C**) that is made of metal as a mounting member is disposed on the front side of the frame **22** (leg plate **22B**). The case **28** is provided with a locking plate **28A** having a substantially rectangular plate shape. The case **28** is mounted to the leg plate **22B** by fixing the upper part and the lower part of the locking plate **28A** to the leg plate **22B** with screws **30**. A locking hole **28C** having a rectangular cross section as a locking part is formed at the lower end part of the locking plate **28A**, and the locking hole **28C** is opened rearward. The locking hole **28C** extends in a downward direction as being directed leftward. Both ends of the locking hole **28C** in the longitudinal direction are opened. A locking cylinder **28B** having a substantially bottomed cylindrical shape is integrally formed on the front side of the locking plate **28A**. The axial direction of the locking cylinder **28B** is parallel to the front-rear direction. The inside of the locking cylinder **28B** penetrates the locking plate **28A** and is opened to the rear side of the locking plate **28A**. The front part of the spool **24** is coaxially inserted into the case **28**.

(29) A torsion shaft **32** (see FIG. **4**) that is made of metal and has a substantially circular columnar shape as a deformation member (energy absorbing member) is coaxially disposed in the case **28**. The front end part of the torsion shaft **32** is locked to the front end part in the locking cylinder **28B** in a relatively non-rotatable manner. The rear end part of the torsion shaft **32** is coupled to the spool **24** in a relatively non-rotatable manner to restrict the rotation of the spool **24**.

(30) A boot **34** (see FIGS. **8A**, **8B**, **9A**, and **9B**) having a substantially rectangular cylindrical shape as a holding member is provided in a range from the buckle body **18** to the frame **22**. The boot **34** is made of soft resin and has flexibility (elasticity). In the boot **34**, the axial direction is set to be the up-down direction, and a dimension in the right-left direction is smaller than a dimension in the front-rear direction. A thickness direction of the boot **34** is set to be the right-left direction. The upper part of the boot **34** gradually becomes smaller downward in the right-left direction. The boot **34** holds the buckle body **18** in a manner such that the lower side part of the buckle body **18** is fitted into the upper part of the boot **34** to restrict the downward movement of the buckle body **18**.

(31) A lower part of the boot **34** is made to serve as an accommodation part **36**. The accommodation part **36** protrudes forward and leftward with respect to a part immediately above the boot **34**. An upper support surface **36B** that constitutes a support target surface **36A** and has a substantially L-shape in bottom view is formed on the upper surfaces of the front part and the left part of the inside of the accommodation part **36**. The upper support surface **36B** is disposed vertically in the up-down direction. A predetermined number (one in the embodiment) of support groove **36C** having a substantially triangular columnar shape is formed in the left side part of the upper surface left part in the accommodation part **36**. The support groove **36C** extends in the front-rear direction. An upper-left inclined surface **36D** as an inclination part constituting the support target surface **36A** is formed on the upper surface (bottom surface) of the support groove **36C**. The upper-left inclined surface **36D** is inclined in a downward direction as being directed to the right (the inner side of the boot **34** in the right-left direction). A support protrusion **36E** having a substantially triangular prism shape is integrally formed at the rear part of the right wall inner surface of the boot **34** immediately above the accommodation part **36**. The support protrusion **36E** protrudes leftward. An upper-right inclined surface **36F** as the inclination part constituting the support target surface **36A** is formed on the lower surface of the support protrusion **36E**. The upper-right inclined surface **36F** is inclined in a downward direction as being directed to the left (the inner side of the boot **34** in the right-left direction).

(32) An exposure hole **36G** is formed to pass through a front wall (side wall) of the accommodation part **36** except for an upper and left corner part and an upper and right corner part. The exposure hole **36G** exposes the inside of the accommodation part **36** to the front side. A covering wall **36H** having a triangular plate shape as a covering part is formed at the upper and left corner part and the

upper and right corner part of the front wall of the accommodation part **36**.

(33) The belt **20** is inserted into the boot **34**, and the frame **22** (including the spool **24**) and the protector **26** are accommodated in the accommodation part **36**. The exposure hole **36G** of the accommodation part **36** exposes the leg plate **22B** of the frame **22** and the front part of the protector **26** to the front side. The case **28** protrudes to the front side from the exposure hole **36G** of the accommodation part **36** (see FIGS. **3** and **4**). The covering wall **36H** on the left side of the accommodation part **36** covers the front side of the leg plate **22B** and the protector **26**, and the covering wall **36H** on the right side of the accommodation part **36** covers the front side of the protector **26**.

(34) The upper support surface **36B**, the upper-left inclined surface **36D**, and the upper-right inclined surface **36F** of the boot **34** are supported from the lower side by the lower support surface **26B**, the lower-left inclined surface **26D**, and the lower-right inclined surface **26F** of the protector **26**, respectively. As a result, the boot **34** is supported from the lower side by the protector **26** and the frame **22**, and the downward movement of the boot **34** is restricted. An elastic contraction force is applied to the boot **34** between the buckle body **18** and the protector **26** in a state where the belt **20** is wound around the spool **24** as described above, and thus the tension is applied to the belt **20**. The boot **34** biases the protector **26** and the frame **22** downward to be self-standing with respect to the protector **26** and the frame **22** and biases the buckle body **18** upward to be self-standing.

(35) A clip **38** that is made of hard resin and has a substantially columnar shape as a fixing member passes through a lower end part of the left wall of the boot **34** (accommodation part **36**) and the back plate **22A** of the frame **22**. A proximal end (left end) of the clip **38** is made in a substantially elliptical plate shape, and a distal end part (right end part) of the clip **38** is formed in a substantially trapezoidal columnar shape. The axial direction of the distal end part of the clip **38** is set to be the front-rear direction, and the distal end part of the clip **38** protrudes upward and downward. After the distal end part of the clip **38** is temporarily elastically reduced in the up-down direction and passes through the left wall of the boot **34** and the back plate **22A** of the frame **22**, the distal end part is caused to pass through the left wall of the boot **34** and the back plate **22A** to be elastically expanded (elastically restored) in the up-down direction. Therefore, the left wall of the boot **34** and the back plate **22A** of the frame **22** are sandwiched between the proximal end and the distal end part of the clip **38**, and the left wall of the boot **34** is fixed to the back plate **22A** of the frame **22** by the clip **38**, whereby tilting of the boot **34** rightward is prevented.

(36) A fixing piece **40** (see FIGS. **5A**, **5B**, and **5C**) having a long plate shape as a fixing part is integrally formed at a front end part of the right wall of the boot **34**, and the fixing piece **40** is extended downward. The fixing piece **40** is bent leftward along the lower and right corner part of the leg plate **22B** of the frame **22** at the proximal end part, and a part on the distal end side of the fixing piece **40** from the proximal end part is extended in the downward direction as being directed leftward.

(37) A hook hole **40A** having an elongated rectangular shape as a hooking target part is formed to penetrate an intermediate part (intermediate part in the longitudinal direction) of the fixing piece **40**, and the hook hole **40A** is extended in the longitudinal direction of the fixing piece **40**. The hook protrusion **22D** of the leg plate **22B** passes through the hook hole **40A**. The left surface of the hook hole **40A** approaches or abuts on a proximal-end side part of the hook protrusion **22D**, and the hook protrusion **22D** is fitted to the hook hole **40A** in the front-rear direction. The distal-end side part of the fixing piece **40** from the hook hole **40A** can abut on the distal end part of the hook protrusion **22D**, and the downward movement of the distal-end side part of the fixing piece **40** from the hook hole **40A** is restricted by the distal end part of the hook protrusion **22D**. As a result, the fixing piece **40** (hook hole **40A**) is hooked on the hook protrusion **22D**.

(38) The intermediate part of the fixing piece **40** is inserted into the locking hole **28C** of the case **28** (locking plate **28A**) in a part including the front side part from the hook hole **40A**, and the lower surface of the locking hole **28C** locks the downward movement of the intermediate part of the

fixing piece **40**. A part of the locking plate **28A** below the locking hole **28C** approaches or abuts on the hook protrusion **22D** of the leg plate **22B**. As a result, leftward tilting of the boot **34** is suppressed by fixing the fixing piece **40** to the frame **22**.

(39) One end of a pair of wire harnesses **42** (wiring, see FIG. 2) is electrically connected to the buckle switch in the buckle body **18**. The wire harness **42** is routed below the boot **34** from the lower end of the buckle body **18** through the inside of the boot **34** (including a space between the rear wall of the boot **34**, and the rear part of the protector **26** and the leg plate **22C** of the frame **22**). One end part of the wire harness **42** is disposed on the right side of the belt **20** above the protector **26** and is bent around the rear part of the protector **26**. The wire harness **42** is provided with two (may be three or more) electric wires **42A**. The electric wire **42A** is made into, for example, an AVSS line (thin low-voltage electric wire for an automobile) and is provided with a sheath on the outer periphery. A vinyl tube **42B** having a circular tubular shape as a covering tube is provided on the other end side from one end part of the pair of wire harnesses **42**, and the vinyl tube **42B** has flexibility. Four electric wires **42A** are inserted into the vinyl tube **42B**, and the four electric wires **42A** are protected by the vinyl tube **42B**. A diameter dimension of the electric wire **42A** is smaller than a dimension H (see FIG. 2) in the up-down direction from the protrusion proximal end of the support column **26E** to the lower end of the lower-right inclined surface **26F** in the protector **26**.

(40) The buckle switch in the buckle body **18** is electrically connected to a control device **44** of the vehicle via the pair of wire harnesses **42**. For example, an airbag device **46** and a warning device **48** of the vehicle are electrically connected to the control device **44**. In a case in which the buckle switch does not detect the engagement of the tongue **16** with the buckle body **18** (the attachment of the webbing **14** to the occupant), the airbag device **46** is not actuated at the time of collision of the vehicle or the warning device **48** warns the occupant to urge the attachment of the webbing **14**, under the control of the control device **44**. In a case in which the buckle switch detects the engagement of the tongue **16** with the buckle body **18** (the attachment of the webbing **14** to the occupant), the airbag device **46** is actuated at the time of collision of the vehicle, and the warning device **48** does not warn the occupant to urge the attachment of the webbing **14**, under the control of the control device **44**.

(41) Next, the action of the embodiment will be described.

(42) In the seat belt device **12** having the above configuration, in the buckle device **10**, the torsion shaft **32** restricts the rotation of the spool **24** to restrict the pull-out of the belt **20** from the spool **24** and the upward extension of the buckle body **18**, and the webbing **14** is attached to the occupant by engaging the tongue **16** of the webbing **14** with the buckle body **18**.

(43) At the time of collision of the vehicle (in emergency), the locking mechanism of the winding device locks the pulling-out of the webbing **14** from the winding device, whereby the occupant is restrained by the webbing **14**. For example, in a case in which an inertial force is applied to the occupant and the webbing **14** is pulled by the occupant, a pulling force from the spool **24** is applied to the belt **20** from the webbing **14** via the tongue **16** and the buckle body **18**, whereby a rotational force is applied to the spool **24**. The torsion shaft **32** is deformed to be twisted (plastically deformed) between the front end part and the rear end part by the rotational force applied to the spool **24**, and the rotation of the spool **24** is allowed, so that the belt **20** is allowed to be pulled out from the spool **24** and the buckle body **18** (including the tongue **16**) is allowed to extend upward. Therefore, a load applied from the webbing **14** to the occupant (particularly, the chest part) is reduced (restricted to a load (force limiter load) twisted deformation of the torsion shaft **32**), and kinetic energy of the occupant is absorbed by twisted deformation of the torsion shaft **32**, whereby the occupant is protected.

(44) The buckle body **18** is held in the upper part of the boot **34**. The frame **22** and the protector **26** are accommodated in the accommodation part **36** below the boot **34**. The upper support surface **36B**, the upper-left inclined surface **36D**, and the upper-right inclined surface **36F** of the support target surface **36A** of the boot **34** are supported from the lower side by the lower support surface

26B, the lower-left inclined surface 26D, and the lower-right inclined surface 26F of the support surface 26A of the protector 26, respectively, and the boot 34 is supported from the lower side by the frame 22 and the protector 26.

(45) Here, the upper-left inclined surface 36D and the upper-right inclined surface 36F of the boot 34 and the lower-left inclined surface 26D and the lower-right inclined surface 26F of the protector 26 are inclined in the downward direction (from the buckle body 18 side to the protector 26 side) as being directed to the inside of the boot 34 in the right-left direction (thickness direction). Therefore, for example, in a case in which a load from the occupant is input to the buckle body 18 and the boot 34 from the upper side, a displacement force toward the inner side of the protector 26 in the right-left direction can be applied to the boot 34 by the upper-left inclined surface 36D and the upper-right inclined surface 36F, and the lower-left inclined surface 26D and the lower-right inclined surface 26F. As a result, it is possible to suppress an occurrence of a situation in which the boot 34 is displaced to the outside of the protector 26 in the right-left direction, and thus the support of the boot 34 by the frame 22 and the protector 26 is released. It is possible to suppress the downward displacement of the boot 34 (to the side of the frame 22 and the protector 26 side) with respect to the frame 22 and the protector 26.

(46) The upper-left inclined surface 36D and the upper-right inclined surface 36F are provided at the boot 34 (support target surface 36A), and the lower-left inclined surface 26D and the lower-right inclined surface 26F are provided at the protector 26 (support surface 26A). Therefore, it is possible to effectively apply a displacement force toward the inner side of the protector 26 in the right-left direction to the boot 34 by the upper-left inclined surface 36D and the upper-right inclined surface 36F, and the lower-left inclined surface 26D and the lower-right inclined surface 26F. It is possible to effectively suppress the downward displacement of the boot 34 with respect to the frame 22 and the protector 26.

(47) Furthermore, the upper-left inclined surface 36D and the upper-right inclined surface 36F of the boot 34 (support target surface 36A) are supported by the lower-left inclined surface 26D and the lower-right inclined surface 26F of the protector 26 (support surface 26A), respectively. Therefore, it is possible to effectively apply a displacement force toward the inner side of the protector 26 in the right-left direction to the boot 34 by the upper-left inclined surface 36D and the upper-right inclined surface 36F, and the lower-left inclined surface 26D and the lower-right inclined surface 26F. It is possible to effectively suppress the downward displacement of the boot 34 with respect to the frame 22 and the protector 26.

(48) In addition, the upper-left inclined surface 36D and the lower-left inclined surface 26D are provided on the left side (one side in the thickness direction of the boot 34) of the boot 34 and the protector 26, and the upper-right inclined surface 36F and the lower-right inclined surface 26F are provided on the right side (the other side in the thickness direction of the boot 34) of the boot 34 and the protector 26. Therefore, it is possible to effectively apply a displacement force toward the inner side of the protector 26 in the right-left direction to the boot 34 by the upper-left inclined surface 36D and the lower-left inclined surface 26D, and the upper-right inclined surface 36F and the lower-right inclined surface 26F. It is possible to effectively suppress the downward displacement of the boot 34 with respect to the frame 22 and the protector 26.

(49) A plurality of lower-left inclined surfaces 26D are provided on the left side of the protector 26 (one side in the thickness direction of the boot 34). Therefore, it is possible to effectively apply the displacement force toward the inner side of the protector 26 in the right-left direction to the boot 34 by the plurality of lower-left inclined surfaces 26D, and it is possible to effectively suppress the downward displacement of the boot 34 with respect to the frame 22 and the protector 26.

(50) The support target surface 36A and the support surface 26A are not provided on the front side and the left side (both sides in the boot 34 circumferential direction) of the upper-right inclined surface 36F and the lower-right inclined surface 26F. Therefore, even in a case in which the support target surface 36A and the support surface 26A are not provided on the front side and the left side

of the upper-right inclined surface 36F and the lower-right inclined surface 26F, it is possible to apply the displacement force toward the inner side of the protector 26 in the right-left direction to the boot 34 by the upper-right inclined surface 36F and the lower-right inclined surface 26F. It is possible to suppress the downward displacement of the boot 34 with respect to the frame 22 and the protector 26.

(51) In addition, the exposure hole 36G of the boot 34 (accommodation part 36) exposes the leg plate 22B of the frame 22 and the protector 26 to the front side, and the covering wall 36H of the accommodation part 36 covers the front side of at least one of the leg plate 22B or the protector 26. Therefore, in a case in which the boot 34 is subjected to the displacement force toward the inner side of the protector 26 in the right-left direction, it is possible to suppress the occurrence of a situation in which the frame 22 and the protector 26 come out from the exposure hole 36G, by the covering wall 36H, to suppress the displacement of the boot 34 to the inner side of the protector 26 in the right-left direction, and to suppress the downward displacement of the boot 34 with respect to the frame 22 and the protector 26.

(52) The diameter dimension of the electric wire 42A of the wire harness 42 is smaller than the dimension H in the up-down direction from the protrusion proximal end of the support column 26E to the lower end of the lower-right inclined surface 26F in the protector 26. Therefore, it is possible to restrict sandwiching of the electric wire 42A between the upper-right inclined surface 36F of the boot 34 and the lower-right inclined surface 26F of the protector 26 by the support column 26E.

(53) In the embodiment, a plurality of inclination parts (lower-left inclined surface 26D) is provided on the left side (one side in the thickness direction of the boot 34) of the protector 26. Along with or instead of this, a plurality of inclination parts may be provided on the left side (one side in the thickness direction of the boot 34) of the boot 34, a plurality of inclination parts may be provided on the right side (one side in the thickness direction of the boot 34) of the protector 26, and a plurality of inclination parts may be provided on the right side (one side in the thickness direction of the boot 34) of the boot 34.

(54) In addition, in the embodiment, the inclination part (the lower-left inclined surface 26D and the upper-left inclined surface 36D) is provided on the left side (one side in the thickness direction of the boot 34) of the boot 34 and the protector 26, and the inclination part (the lower-right inclined surface 26F and the upper-right inclined surface 36F) is provided on the right side (the other side in the thickness direction of the boot 34) of the boot 34 and the protector 26. The inclination part may be provided on the left side or the right side (one side in the thickness direction of the boot 34) of the boot 34 and the protector 26.

(55) Further, in the embodiment, the boot 34 (support target surface 36A) is provided with the inclination part (the upper-left inclined surface 36D and the upper-right inclined surface 36F), and the protector 26 (support surface 26A) is provided with the inclination part (the lower-left inclined surface 26D and the lower-right inclined surface 26F). The inclination part only has to be provided on at least one of the boot 34 (support target surface 36A) or the protector 26 (support surface 26A).

(56) In the embodiment, the spool 24 is provided in the frame 22. The belt 20 may be coupled to the frame 22 without providing the spool 24 in the frame 22. In this case, the torsion shaft 32 does not need to be provided.

(57) In the embodiment, the deformation member is the torsion shaft 32. The deformation member may be a member other than the torsion shaft 32.

(58) In the embodiment, four electric wires 42A are provided. One or more electric wires 42A only has to be provided.

Claims

1. A buckle device comprising: a buckle body configured to engage with a tongue provided at a webbing attached to an occupant; a holding member that holds the buckle body at an internal portion and is provided with a support target surface at the internal portion; a support that is provided with a support surface and supports the holding member in a manner such that the support surface supports the support target surface; and an inclination part that is provided at at least one of the support target surface or the support surface and is inclined in a direction from a buckle body side toward a support side as being directed inward in a holding-member thickness direction.
 2. The buckle device according to claim 1, wherein the inclination part is provided at the support target surface and the support surface.
 3. The buckle device according to claim 2, wherein the inclination part at the support surface supports the inclination part at the support target surface.
 4. The buckle device according to claim 1, wherein the inclination part is provided on both sides in the holding-member thickness direction.
 5. The buckle device according to claim 1, wherein a plurality of the inclination parts are provided on one side in the holding-member thickness direction.
 6. The buckle device according to claim 1, wherein the support target surface and the support surface are not provided on both sides of the inclination part in a holding-member circumferential direction.
 7. The buckle device according to claim 1, further comprising: an accommodation part that is provided at the holding member and accommodates the support; an exposure hole that is provided at a side wall of the accommodation part and exposes the support; and a covering part that is provided at the side wall and covers the support.
 8. The buckle device according to claim 1, wherein the holding member has flexibility.
 9. The buckle device according to claim 1, wherein the holding member is fixed to the support.
 10. A buckle device comprising: a buckle body configured to engage with a tongue provided at a webbing attached to an occupant; a holding member that holds the buckle body and is provided with an accommodation part; a support that is accommodated in the accommodation part from a side opposite to the buckle body and supports the holding member; an exposure hole that is provided at a side wall of the accommodation part and exposes the support; and a covering part that is provided at the side wall and covers the support.
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